

Canonical Positive Symplectic Diagonalization of the Pais–Uhlenbeck Oscillator

Abstract

We prove that the unequal-frequency PT-symmetric Pais–Uhlenbeck oscillator admits a canonical positive complex-symplectic diagonalization, and that the metaplectic logarithm of the unique Hermitian positive-definite diagonalizer is exactly the Bender–Mannheim generator $Q = \alpha pq + \beta xy$. The generator is thereby *reconstructed* from the normal-form data (G, J, G_0) rather than assumed. We determine the full solution family of the diagonalization problem: its stabilizer is $SO(2, \mathbb{C}) \times SO(2, \mathbb{C})$, of which only a finite subgroup is unitary in the canonical coordinates, so the positive polar factor is not an invariant of the problem, while the Hermitian positive-definite diagonalizer is unique. We classify all positive metric operators compatible with pseudo-Hermiticity, compute the observable-space metric spectrum $\{e^r, e^r, e^{-r}, e^{-r}\}$ with $r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}$, and show that its affine-invariant distance from the identity is exactly $2r$. In the equal-frequency limit this distance diverges logarithmically, and we prove that the limiting Jordan block admits no positive-definite invariant Hermitian form. All finite-dimensional algebraic identities appearing in this paper have been independently verified symbolically, numerically, and by machine-checked Lean 4 proofs.

1 Introduction

The Pais–Uhlenbeck (PU) oscillator [1] is the minimal higher-derivative model: a single degree of freedom governed by the fourth-order Lagrangian

$$L = \frac{\gamma}{2} (\ddot{z}^2 - (\omega_1^2 + \omega_2^2) \dot{z}^2 + \omega_1^2 \omega_2^2 z^2), \quad \gamma > 0, \quad \omega_1 > \omega_2 > 0. \quad (1)$$

Its Ostrogradsky Hamiltonian is unbounded below, and canonical quantization on the naive inner product produces negative-norm states. This ghost problem is the standard obstruction to higher-derivative theories, including conformal gravity, where the same mode structure appears [5].

Bender and Mannheim [3, 4] observed that the Ostrogradsky Hamiltonian, continued to a PT-symmetric operator H_{PT} , has real positive spectrum, and that an operator $Q = \alpha pq + \beta xy$ with explicit coefficients intertwines H_{PT} with a manifestly positive two-oscillator Hamiltonian: $e^{-Q/2} H_{\text{PT}} e^{Q/2}$ is Hermitian, and $\eta = e^{-Q}$ is a positive metric restoring unitarity. Their derivation proceeds by an operator ansatz whose coefficients are fixed by demanding that the transformed Hamiltonian be Hermitian. Related constructions eliminate the PU ghost by complex canonical transformations [10] or by an alternative Hamiltonian formulation [9]; the general framework of pseudo-Hermitian quantum mechanics is due to Scholtz, Geyer and Hahne [6] and Mostafazadeh [7, 8].

This paper treats the problem as one of *finite-dimensional symplectic geometry* and proves the following.

1. The PT Hamiltonian is a complex quadratic form $H_{\text{PT}} = \frac{1}{2}V^\top GV$ with a complex-symmetric (not Hermitian) matrix G , and the diagonalization problem

$$S^\top JS = J, \quad S^\top GS = G_0 \quad (2)$$

with G_0 the positive two-oscillator normal form, has an explicit solution family.

2. After a canonical symplectic normalization (Section 3), the problem (2) has a *unique* Hermitian positive-definite solution $S_+ \in \text{Sp}(4, \mathbb{C})$. Uniqueness requires no genericity assumption beyond $\omega_1 > \omega_2 > 0$.
3. The metaplectic logarithm of S_+ is exactly the Bender–Mannheim generator: the quadratic operator $\hat{Q}$ with $e^{-\hat{Q}/2} V e^{\hat{Q}/2} = S_+ V$ is $\hat{Q} = \alpha pq + \beta xy$ with the Bender–Mannheim coefficients. The generator is thus reconstructed from (G, J, G_0) and positivity alone (Section 4); no ansatz enters.
4. The stabilizer of the normal form is $SO(2, \mathbb{C}) \times SO(2, \mathbb{C})$ (Section 3). Only a finite subgroup of it is unitary in the canonical coordinates, and consequently the positive polar factor of a general solution of (2) is *not* unique; the correct uniqueness statement concerns the Hermitian positive-definite solution.
5. All positive metric operators η' compatible with pseudo-Hermiticity of H_{PT} form the family $\eta' = \rho^\dagger W \rho$ with $W > 0$ commuting with the diagonalized Hamiltonian; positive metrics are not unique (Section 5).
6. The observable-space metric $M_{\text{obs}} = S_+^\dagger S_+$ has spectrum $\{e^r, e^r, e^{-r}, e^{-r}\}$ with $r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}$, and its affine-invariant distance from the identity is exactly

$$d(I, M_{\text{obs}}) = 2r = 2 \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2} \quad (3)$$

(Section 6). For $\omega_{1,2} = \omega \pm \epsilon$ one has $r = \log(\omega/\epsilon)$ exactly, so the equal-frequency exceptional point lies at infinite affine-invariant distance, and no scalar renormalization of M_{obs} has a finite positive nondegenerate limit.

7. At the exceptional point the flow develops a Jordan block, and we prove that a Jordan block admits no positive-definite invariant Hermitian form: nondegenerate invariant forms are indefinite and positive-semidefinite ones are degenerate (Section 7).

Throughout, we distinguish the observable-space (phase-space) geometry of the 4×4 matrices from the Hilbert-space geometry of the unbounded metric operator $\eta = e^{-Q}$; the two are related by the metaplectic representation but must not be identified (Remark 6.4).

Verification. All finite-dimensional algebraic identities appearing in this paper have been independently verified symbolically (SymPy), numerically (50–80 significant digits, including a near-degenerate frequency pair), and—for the identities $K^2 = -r^2 I$, $S^\top JS = J$, $\det S = 1$, $S^\top GS = G_0$, and the complete Jordan-block obstruction of Section 7—by machine-checked Lean 4 proofs against Mathlib, with no unproved placeholders. The accompanying repository contains the complete verification pipeline. Operator-theoretic statements about the unbounded metric η are made at the formal algebraic level on the oscillator domain and are flagged as such.

2 Quadratic Hamiltonian formulation

2.1 Conventions

We collect the canonical operators in the column vector $V = (x, y, p, q)^\top$ with

$$[x, p] = i, \quad [y, q] = i, \quad (4)$$

all other basic commutators vanishing; equivalently $[V_a, V_b] = iJ_{ab}$ with the symplectic matrix

$$J = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad J^\top = -J, \quad J^2 = -I. \quad (5)$$

A quadratic Hamiltonian $H = \frac{1}{2}V^\top GV$ with $G = G^\top$ generates the linear flow $\dot{V} = AV$, $A = JG$. The same formulas define the classical theory with $\{V_a, V_b\} = J_{ab}$; all matrix identities below are representation-independent.

2.2 The PT Hamiltonian

Introducing the Ostrogradsky variables $x = \dot{z}$ and their momenta, the Lagrangian (1) gives $H_{\text{PU}} = p^2/(2\gamma) + p_z x + \frac{\gamma}{2}(\omega_1^2 + \omega_2^2)x^2 - \frac{\gamma}{2}\omega_1^2\omega_2^2 z^2$, whose $-z^2$ term encodes the Ostrogradsky instability. The PT reformulation [3, 4] substitutes $z = iy$, $p_z = -iq$ (preserving $[y, q] = i$) and yields

$$H_{\text{PT}} = \frac{p^2}{2\gamma} - iqx + \frac{\gamma}{2}(\omega_1^2 + \omega_2^2)x^2 + \frac{\gamma}{2}\omega_1^2\omega_2^2 y^2 = \frac{1}{2}V^\top GV, \quad (6)$$

$$G = \begin{pmatrix} \gamma(\omega_1^2 + \omega_2^2) & 0 & 0 & -i \\ 0 & \gamma\omega_1^2\omega_2^2 & 0 & 0 \\ 0 & 0 & \gamma^{-1} & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}, \quad (7)$$

where the cross term uses $[x, q] = 0$, so no ordering ambiguity arises. The matrix G is complex symmetric but not Hermitian: H_{PT} is PT-symmetric, not Hermitian.

Proposition 2.1. *The flow matrix $A = JG$ has characteristic polynomial*

$$\det(\lambda I - A) = (\lambda^2 + \omega_1^2)(\lambda^2 + \omega_2^2), \quad (8)$$

computed directly from the determinant. For $\omega_1 \neq \omega_2$ the eigenvalues $\pm i\omega_1, \pm i\omega_2$ are distinct and purely imaginary.

The target normal form is the positive two-oscillator Hamiltonian

$$\tilde{H} = \frac{p^2}{2\gamma} + \frac{q^2}{2\gamma\omega_1^2} + \frac{\gamma\omega_1^2}{2}x^2 + \frac{\gamma\omega_1^2\omega_2^2}{2}y^2 = \frac{1}{2}V^\top G_0 V, \quad G_0 = \text{diag}(\gamma\omega_1^2, \gamma\omega_1^2\omega_2^2, \gamma^{-1}, (\gamma\omega_1^2)^{-1}), \quad (9)$$

whose modes (x, p) and (y, q) oscillate with frequencies ω_1 and ω_2 respectively. Since G is complex, the classical Williamson theorem [16] for real positive forms does not apply; the content of Sections 3–4 is a positive symplectic normal form over $\mathbb{C}$ for this PT class, together with a uniqueness theorem.

3 Canonical positive symplectic diagonalization

3.1 The parameters

Throughout, set

$$r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2} > 0, \quad \sigma = \sqrt{\omega_1^2 - \omega_2^2} > 0. \quad (10)$$

Exponentiating $r/2$ gives the hyperbolic values used repeatedly below:

$$\cosh \frac{r}{2} = \frac{\omega_1}{\sigma}, \quad \sinh \frac{r}{2} = \frac{\omega_2}{\sigma}, \quad \cosh r = \frac{\omega_1^2 + \omega_2^2}{\omega_1^2 - \omega_2^2}, \quad \sinh r = \frac{2\omega_1\omega_2}{\omega_1^2 - \omega_2^2}. \quad (11)$$

3.2 Canonical normalization

For $d_x, d_y > 0$ the diagonal matrix $D = \text{diag}(d_x, d_y, d_x^{-1}, d_y^{-1})$ satisfies $D^\top J D = J$: it is a real canonical rescaling. We normalize the coordinates by

$$V' = DV, \quad d_x d_y = \gamma\omega_1\omega_2, \quad (12)$$

and write $G' = D^{-\top} G D^{-1}$, $G'_0 = D^{-\top} G_0 D^{-1}$ for the quadratic forms in the normalized coordinates. Only the product $d_x d_y$ is fixed; every statement below is invariant under the residual split d_x/d_y , which acts by conjugation with a positive diagonal symplectic matrix. For the symmetric choice $d_x = d_y = \sqrt{\gamma\omega_1\omega_2}$,

$$G' = \begin{pmatrix} \frac{\omega_1^2 + \omega_2^2}{\omega_1\omega_2} & 0 & 0 & -i \\ 0 & \omega_1\omega_2 & 0 & 0 \\ 0 & 0 & \omega_1\omega_2 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}, \quad G'_0 = \text{diag}(t, \omega_1\omega_2, \omega_1\omega_2, t^{-1}), \quad t = \frac{\omega_1}{\omega_2}. \quad (13)$$

Both G' and G'_0 are block diagonal with respect to the planes $\Pi_{xq} = \text{span}\{x, q\}$ and $\Pi_{yp} = \text{span}\{y, p\}$.

The reason for the condition $d_x d_y = \gamma\omega_1\omega_2$ is Proposition 3.3 below: it is exactly the normalization that renders the diagonalizer Hermitian.

3.3 Existence

Definition 3.1. Let $B \in M_4(\mathbb{C})$ be given, in the ordering (x, y, p, q) , by

$$B = \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}, \quad B^\dagger = B, \quad B^2 = I, \quad (14)$$

a Hermitian involution with two-dimensional eigenspaces $\ker(B \mp I)$, and set

$$S_+ = \cosh \frac{r}{2} I + \sinh \frac{r}{2} B = \exp\left(\frac{r}{2} B\right) = \frac{tI + B}{\sqrt{t^2 - 1}}. \quad (15)$$

The last equality follows from (11). Since B is a Hermitian involution, S_+ is Hermitian with spectrum $\{e^{r/2}, e^{r/2}, e^{-r/2}, e^{-r/2}\}$; in particular $S_+ > 0$ and $\det S_+ = 1$.

Theorem 3.2 (Existence). *The matrix S_+ of (15) satisfies*

$$S_+^\top J S_+ = J, \quad S_+^\top G' S_+ = G'_0. \quad (16)$$

Proof. $B^\top J = -JB$ by inspection, so B is J -Hamiltonian and $e^{(r/2)B}$ is symplectic: $S_+^\top J S_+ = (\cosh^2 \frac{r}{2} - \sinh^2 \frac{r}{2})J + \cosh \frac{r}{2} \sinh \frac{r}{2} (B^\top J + JB) = J$, using $B^\top JB = -JB^2 = -J$.

For the congruence, note that S_+ preserves the splitting $\Pi_{xq} \oplus \Pi_{yp}$, acting on each plane (in the bases (x, q) and (y, p) , with $c = \cosh \frac{r}{2}$, $s = \sinh \frac{r}{2}$) by $\begin{pmatrix} c & is \\ -is & c \end{pmatrix}$, while by (13) G' restricts to $\begin{pmatrix} t+t^{-1} & -i \\ -i & 0 \end{pmatrix}$ on Π_{xq} and to $\omega_1 \omega_2 I_2$ on Π_{yp} . On Π_{yp} , $\begin{pmatrix} c & -is \\ is & c \end{pmatrix} \begin{pmatrix} c & is \\ -is & c \end{pmatrix} = (c^2 - s^2)I_2 = I_2$, matching $G'_0|_{\Pi_{yp}} = \omega_1 \omega_2 I_2$. On Π_{xq} , using $c = \omega_1/\sigma$, $s = \omega_2/\sigma$,

$$\begin{pmatrix} c & -is \\ is & c \end{pmatrix} \begin{pmatrix} t+t^{-1} & -i \\ -i & 0 \end{pmatrix} \begin{pmatrix} c & is \\ -is & c \end{pmatrix} = \begin{pmatrix} (t+t^{-1})c^2 - 2cs & i((t+t^{-1})cs - c^2 - s^2) \\ -i((t+t^{-1})cs - c^2 - s^2) & -(t+t^{-1})s^2 + 2cs \end{pmatrix} = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix},$$

where the off-diagonal entries vanish because $(t+t^{-1})cs = (\omega_1^2 + \omega_2^2)/\sigma^2 = c^2 + s^2$, and the diagonal entries reduce to ω_1/ω_2 and ω_2/ω_1 by direct simplification. This is G'_0 restricted to the two planes. $\square$

Undoing the normalization, $S = D^{-1}S_+D$ solves the original problem $S^\top JS = J$, $S^\top GS = G_0$; the congruence, symplecticity and $\det S = 1$ in the original coordinates are among the Lean-verified statements. The flow relation $S^{-1}AS = A_0 = JG_0$ follows from the congruence by $JS^\top = S^{-1}J$.

Proposition 3.3 (Hermiticity requires the normalization). *Let $D = \text{diag}(d_x, d_y, d_x^{-1}, d_y^{-1})$ with $d_x, d_y > 0$ and let $S = D^{-1}S_+D$ be the induced diagonalizer of (G, J, G_0) . Then S is Hermitian if and only if $d_x d_y = \gamma \omega_1 \omega_2$. In particular the diagonalizer is not Hermitian in the original coordinates unless $\gamma \omega_1 \omega_2 = 1$.*

Proof. Conjugating B by D scales the entries B_{xq}, B_{yp} by $d_x d_y / (\gamma \omega_1 \omega_2)$ relative to the normalized form and the entries B_{py}, B_{qx} by the inverse factor; Hermiticity of the conjugated involution, hence of S , holds iff the factor is 1. $\square$

3.4 The solution family and uniqueness

We now determine *all* solutions of the diagonalization problem in the normalized coordinates. Write $A'_0 = JG'_0$ and let P_1, P_2 be the coordinate projections onto the modes (x, p) and (y, q) of G'_0 , and

$$X_1 = \frac{P_1 A'_0 P_1}{\omega_1}, \quad X_2 = \frac{P_2 A'_0 P_2}{\omega_2}, \quad X_j^2 = -P_j. \quad (17)$$

Explicitly X_1 acts on (x, p) by $\begin{pmatrix} 0 & \lambda_1 \\ -\lambda_1^{-1} & 0 \end{pmatrix}$ and X_2 on (y, q) by $\begin{pmatrix} 0 & \lambda_2^{-1} \\ -\lambda_2 & 0 \end{pmatrix}$, where $\lambda_1, \lambda_2 > 0$ are the *dimensionless mode-aspect parameters* of the chosen canonical coordinates: with $\hbar = 1$ and the symmetric split $d_x = d_y$ they take the values $\lambda_1 = \omega_2$, $\lambda_2 = \omega_1$ in normalized units, and a residual split d_x/d_y rescales them individually while preserving the split-invariant ratio $\lambda_1/\lambda_2 = \omega_2/\omega_1 < 1$. All exceptional conditions below are alignments $\lambda_j = 1$ of a mode aspect with the canonical split — comparisons of dimensionless quantities, not of frequencies with the number 1.

Lemma 3.4 (Stabilizer). *For $\omega_1 \neq \omega_2$,*

$$\text{Stab}(J, G'_0) = \{C : C^\top J C = J, C^\top G'_0 C = G'_0\} = \{C(\theta_1, \theta_2) = \sum_{j=1,2} (\cos \theta_j P_j + \sin \theta_j X_j) : \theta_j \in \mathbb{C}\} \cong SO(2, \mathbb{C}) \quad (18)$$

Proof. ($\supseteq$) On each mode, $X_j^\top J + J X_j = 0$ and $X_j^\top G'_0 + G'_0 X_j = 0$ by inspection, and $X_j^2 = -P_j$, so for $C_j = \cos \theta_j P_j + \sin \theta_j X_j$ both quadratic forms are preserved: $C_j^\top Q C_j = (\cos^2 \theta_j + \sin^2 \theta_j) Q = Q$ on the mode, for $Q \in \{J, G'_0\}$.

($\subseteq$) If C preserves both forms then, using $C^{-\top} = J C J^{-1}$ from symplecticity, $A'_0 C = J G'_0 C = J C^{-\top} G'_0 = C J G'_0 = C A'_0$: C commutes with A'_0 . Since A'_0 has the four distinct eigenvalues $\pm i\omega_1, \pm i\omega_2$ (Proposition 2.1), its commutant is the span of $\{P_1, X_1, P_2, X_2\}$. Writing $C = \sum_j (a_j P_j + b_j X_j)$, the condition $C^\top J C = J$ restricts on each mode to $a_j^2 + b_j^2 = 1$, which is the $SO(2, \mathbb{C})$ relation; $C^\top G'_0 C = G'_0$ is then automatic as in ($\supseteq$). $\square$

Consequently the full solution family of (2) in normalized coordinates is the coset

$$\{S : S^\top J S = J, S^\top G'_0 S = G'_0\} = S_+ \cdot \text{Stab}(J, G'_0). \quad (19)$$

(If S is any solution, $C = S_+^{-1} S$ preserves both forms.)

Theorem 3.5 (Uniqueness). *For every $\gamma > 0$ and $\omega_1 > \omega_2 > 0$, the matrix S_+ is the unique Hermitian positive-definite solution of $S^\top J S = J$, $S^\top G'_0 S = G'_0$.*

Proof. By (19) any solution is $S_+ C(\theta_1, \theta_2)$. Writing out the 16 entries of $S_+ C$ ($c = \cosh \frac{r}{2}$, $s = \sinh \frac{r}{2}$), the Hermiticity conditions $(S_+ C)^\dagger = S_+ C$ include, from the entry pairs $(x, p)/(p, x)$ and $(y, q)/(q, y)$,

$$\omega_2^2 \sin \theta_1 = -\overline{\sin \theta_1}, \quad \sin \theta_2 = -\omega_1^2 \overline{\sin \theta_2}. \quad (20)$$

Taking absolute values, $\lambda_1^2 |\sin \theta_1| = |\sin \theta_1|$ and $|\sin \theta_2| = \lambda_2^2 |\sin \theta_2|$; hence $\sin \theta_1 = 0$ unless $\lambda_1 = 1$ and $\sin \theta_2 = 0$ unless $\lambda_2 = 1$, and these two exceptional alignments cannot occur simultaneously since $\lambda_1/\lambda_2 = \omega_2/\omega_1 < 1$ is split-invariant. If $\lambda_1 = 1$, the (y, q) condition still gives $\sin \theta_2 = 0$, and the entry pair $(x, y)/(y, x)$ gives $\lambda_1 \lambda_2 \sin \theta_2 = -\overline{\sin \theta_1}$, forcing $\sin \theta_1 = 0$ as well; the case $\lambda_2 = 1$ is symmetric. So in all cases $\sin \theta_1 = \sin \theta_2 = 0$, whence $\cos^2 \theta_j = 1$ and $C = \varepsilon_1 P_1 + \varepsilon_2 P_2$ with $\varepsilon_j = \pm 1$. The entry pair $(x, q)/(q, x)$ gives $\cos \theta_2 = \overline{\cos \theta_1}$, i.e. $\varepsilon_1 = \varepsilon_2$, so $C = \pm I$. Finally $S_+(-I) = -S_+ < 0$ is excluded by positivity, leaving $C = I$. $\square$

Proposition 3.6 (Polar factors are not invariant). *Let $S = S_+C$ with $C \in \text{Stab}(J, G'_0)$, and let $S = PU$ be its polar decomposition ($P > 0$, U unitary). Then $P = S_+$ if and only if C is unitary. In the normalized coordinates the unitary elements of $\text{Stab}(J, G'_0)$ form, for $\lambda_1, \lambda_2 \neq 1$, the finite group $\{\pm P_1 \pm P_2\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Consequently the positive polar factor varies over a two-complex-parameter family as C ranges over the stabilizer: it is not an invariant of the diagonalization problem.*

Proof. If C is unitary, S_+C is already a polar decomposition, and by its uniqueness $P = S_+$. Conversely $P = S_+$ forces $U = S_+^{-1}S = C$ unitary. For the second claim, $C(\theta)$ restricted to mode 1 is $\begin{pmatrix} a & b\lambda_1 \\ -b/\lambda_1 & a \end{pmatrix}$ with $a^2 + b^2 = 1$; unitarity requires $|a|^2 + |b|^2\omega_2^{\mp 2} = 1$ for both signs, hence $b = 0$ for $\lambda_1 \neq 1$, and then $|a| = 1$ with $a^2 = 1$, i.e. $a = \pm 1$; likewise on mode 2. An explicit non-unitary example is $C(i\tau, 0)$, $\tau \in \mathbb{R} \setminus \{0\}$, for which $(S_+C)^\dagger(S_+C) \neq S_+^2$. $\square$

Remark 3.7. The stabilizer is $SO(2, \mathbb{C})^2 \cong (\mathbb{C}^\times)^2$ as an abstract group, whose maximal compact subgroup is $U(1)^2$; but its unitary part in the canonical coordinates is generically only $\mathbb{Z}_2^2$, because the $U(1)$ directions of the two $SO(2, \mathbb{C})$ factors cannot be made simultaneously unitary in coordinates where S_+ is Hermitian. The apparent alignment “ $\lambda_j = 1$ ” in the proofs above depends on the residual split freedom d_x/d_y of (12), which can move but not remove the exceptional locus (the ratio λ_1/λ_2 is split-invariant); Theorem 3.5 holds for all parameters.

4 Reconstruction of the Bender–Mannheim generator

We now derive the generator from S_+ ; at no point is the form of Q assumed.

4.1 Quadratic generators and the metaplectic dictionary

For a symmetric matrix $M = M^\top \in M_4(\mathbb{C})$ let $\widehat{Q} = \frac{1}{2}V^\top MV$ and $K = JM$. From $[V_a, V_b] = iJ_{ab}$,

$$[\widehat{Q}, V] = \frac{1}{2}M_{bc}(V_b[V_c, V_a] + [V_b, V_a]V_c)_a = -iKV, \quad (21)$$

componentwise on the column vector V ; the sign is fixed by the convention $[x, p] = +i$. Since $\text{ad}_{\widehat{Q}}$ acts linearly on V , the Baker–Campbell–Hausdorff series sums exactly:

$$e^{-\widehat{Q}/2} V e^{\widehat{Q}/2} = e^{iK/2} V, \quad e^{\widehat{Q}/2} V e^{-\widehat{Q}/2} = e^{-iK/2} V. \quad (22)$$

Thus quadratic operators implement one-parameter symplectic groups: this is the (formal) metaplectic dictionary $\widehat{Q} \leftrightarrow K = JM$ between quadratic generators and the Lie algebra $\mathfrak{sp}(4, \mathbb{C})$ [21].

4.2 The logarithm of S_+

By Definition 3.1, $S_+ = e^{(r/2)B}$, and since B is an involution this is the exact logarithm delivered by the spectral calculus:

$$\log S_+ = \frac{r}{2} B. \quad (23)$$

To realize $S_+ = e^{iK'/2}$ in the dictionary (22) we need $iK'/2 = \log S_+$, i.e.

$$K' = -irB, \quad M' = J^{-1}K' = -JK' = r \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (\text{normalized coordinates}), \quad (24)$$

which is symmetric, as required. Pulling back along the normalization (12), $M = D^\top M' D$ depends only on the product $d_x d_y = \gamma \omega_1 \omega_2$ and equals

$$M = \begin{pmatrix} 0 & \beta & 0 & 0 \\ \beta & 0 & 0 & 0 \\ 0 & 0 & 0 & \alpha \\ 0 & 0 & \alpha & 0 \end{pmatrix}, \quad \alpha = \frac{r}{\gamma \omega_1 \omega_2}, \quad \beta = r \gamma \omega_1 \omega_2 = \alpha \gamma^2 \omega_1^2 \omega_2^2. \quad (25)$$

Theorem 4.1 (Reconstruction). *The quadratic operator associated with the metaplectic logarithm of the unique Hermitian positive-definite diagonalizer is*

$$Q = \frac{1}{2} V^\top M V = \alpha p q + \beta x y, \quad \alpha = \frac{1}{\gamma \omega_1 \omega_2} \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}, \quad \beta = \alpha \gamma^2 \omega_1^2 \omega_2^2, \quad (26)$$

i.e. exactly the Bender–Mannheim generator [4], with $r = \sqrt{\alpha \beta}$ (positive root, since $\alpha, \beta > 0$ for $\omega_1 > \omega_2 > 0$). Moreover $K = JM$ satisfies $K^2 = -r^2 I$, so the exponential closes in the first two powers of K ,

$$S = e^{iK/2} = \cosh \frac{r}{2} I + i \frac{\sinh \frac{r}{2}}{r} K, \quad (27)$$

and $e^{-Q/2} H_{\text{PT}} e^{Q/2} = \frac{1}{2} V^\top (S^\top G S) V = \frac{1}{2} V^\top G_0 V = \tilde{H}$ is the positive two-oscillator Hamiltonian (9).

Proof. The pullback computation above gives M , hence Q ; the identities $K^2 = -r^2 I$ and $S^\top G S = G_0$ are verified directly (and are machine-checked); the last display combines (22) with the congruence. $\square$

The logical order of the construction is therefore

$$(G, J, G_0) \implies S_+ \text{ (existence and uniqueness)} \implies Q \text{ (metaplectic logarithm)}, \quad (28)$$

with positivity of S_+ (after the canonical normalization (12)) as the only selection principle. Beyond H_{PT} itself, the inputs are the choice of the target normal form G_0 and the positivity requirement; nothing else.

5 Pseudo-Hermiticity and metric classification

In this section, operator identities are formal-algebraic: they hold on the common invariant domain of finite linear combinations of the $\tilde{H}$ -oscillator eigenstates, on which all operators below act; Q is essentially self-adjoint there but $e^{\pm Q/2}$ are unbounded, and we make no claims about closures or operator-theoretic similarity. The finite-dimensional identities backing each step are exact and verified.

Since $\alpha, \beta \in \mathbb{R}$ and $[p, q] = [x, y] = 0$, the generator satisfies $Q^\dagger = Q$; set

$$\rho = e^{-Q/2} = \rho^\dagger, \quad \eta = \rho^\dagger \rho = e^{-Q}. \quad (29)$$

Proposition 5.1 (Pseudo-Hermiticity). *Formally, $H_{\text{PT}}^\dagger \eta = \eta H_{\text{PT}}$. At the matrix level this is equivalent to the exact identity*

$$(S^2)^\top G S^2 = \overline{G}, \quad (30)$$

which holds for $S = e^{iK/2}$.

Proof. By (22) with parameter doubled, $e^{-Q} V e^Q = S^2 V$, so $\eta H_{\text{PT}} \eta^{-1} = \frac{1}{2} V^\top ((S^2)^\top G S^2) V$. Since x, y, p, q are Hermitian and G is symmetric, $H_{\text{PT}}^\dagger = \frac{1}{2} V^\top \overline{G} V$. The claim is then (30), verified exactly. Equivalently: with $h = \rho H_{\text{PT}} \rho^{-1} = \tilde{H}$ Hermitian (Theorem 4.1) and $\rho^\dagger = \rho$, the free-algebra identity $\rho (h^\dagger - h) \rho = H_{\text{PT}}^\dagger \eta - \eta H_{\text{PT}}$ gives the statement. $\square$

Theorem 5.2 (Classification of positive metrics). *Let $h = \rho H_{\text{PT}} \rho^{-1} = \tilde{H}$. The map $W \mapsto \eta' = \rho^\dagger W \rho$ is a bijection from*

$$\{W > 0 \text{ invertible}, [W, h] = 0\} \quad \text{onto} \quad \{\eta' > 0 \text{ invertible}, H_{\text{PT}}^\dagger \eta' = \eta' H_{\text{PT}}\}, \quad (31)$$

with inverse $\eta' \mapsto W = \rho^{-\dagger} \eta' \rho^{-1}$, again at the formal level. In particular positive metrics are not unique: $W = I$ gives the canonical $\eta = e^{-Q}$, and any positive invertible function $W = f(N_1, N_2)$ of the two commuting number operators of $\tilde{H}$ gives another metric. For ω_1/ω_2 irrational the commutant of h is generated by N_1, N_2 , so this family is exhaustive.

Proof. With $\rho^\dagger = \rho$ and $H_{\text{PT}} = \rho^{-1} h \rho$, $H_{\text{PT}}^\dagger = \rho h \rho^{-1}$ (using $h^\dagger = h$), a two-line computation gives $H_{\text{PT}}^\dagger \eta' - \eta' H_{\text{PT}} = \rho [h, W] \rho$ for $\eta' = \rho W \rho$: the intertwining relation holds iff $[W, h] = 0$. Positivity and invertibility transfer both ways because congruence by the invertible ρ preserves them, and the two maps are mutually inverse by construction. $\square$

Uniqueness of the metric therefore requires a selection principle beyond positivity and pseudo-Hermiticity. The results of Section 3 identify one: within the symplectic-Gaussian class, demanding that the metric arise from a Hermitian positive-definite *symplectic* diagonalizer selects $\eta = e^{-Q}$ uniquely (Theorem 3.5); the residual W -freedom of Theorem 5.2 is the Hilbert-space counterpart of the stabilizer freedom (19), and the non-invariance of the polar factor (Proposition 3.6) is its finite-dimensional shadow.

6 Geometry of the positive metric cone

6.1 The observable-space metric

Definition 6.1. The observable-space metric of the diagonalization is $M_{\text{obs}} = S_+^\dagger S_+ \in M_4(\mathbb{C})$, a positive-definite Hermitian matrix on the normalized phase-space coordinates.

Since S_+ is Hermitian, $M_{\text{obs}} = S_+^2 = e^{rB} = \cosh r I + \sinh r B$, and B is an involution with two-dimensional eigenspaces, so with (11):

Theorem 6.2. $\text{spec } M_{\text{obs}} = \{e^r, e^r, e^{-r}, e^{-r}\}$, i.e.

$$M_{\text{obs}} = e^r \Pi_+ + e^{-r} \Pi_-, \quad \Pi_\pm = \frac{1}{2}(I \pm B). \quad (32)$$

This holds in the canonically normalized coordinates (12); in the original coordinates $S^\dagger S$ has a different, γ -dependent spectrum.

On the cone of positive-definite Hermitian matrices we use the affine-invariant Riemannian metric $\langle U, V \rangle_G = \text{tr}(G^{-1}UG^{-1}V)$ (no normalization factor), whose geodesic distance from the identity is $d(I, M) = \|\log M\|_F$ [17].

Theorem 6.3 (Distance).

$$d(I, M_{\text{obs}}) = \|\log M_{\text{obs}}\|_F = \|rB\|_F = 2r = 2 \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}. \quad (33)$$

Proof. $\log M_{\text{obs}} = rB$ by the spectral calculus, and $\|rB\|_F = r\sqrt{\text{tr } B^\dagger B} = r\sqrt{\text{tr } I} = 2r$. $\square$

Remark 6.4 (Observable space versus Hilbert space). $M_{\text{obs}} = S_+^\dagger S_+$ and $\eta = e^{-Q}$ must not be identified as operators. M_{obs} is a 4×4 matrix acting on phase-space labels: it measures how the canonical coordinates are deformed by the diagonalization, and its spectrum $\{e^{\pm r}\}$ is the squeezing content of S_+ . The metric $\eta = e^{-Q}$ is an unbounded positive operator on Hilbert space with purely continuous spectrum $(0, \infty)$ (the spectrum of Q is $\mathbb{R}$, since Q is unitarily equivalent to a generator of dilations). They are related, not equal: $\rho = e^{-Q/2}$ is the formal metaplectic implementer of S_+ via (22), and $\eta = \rho^\dagger \rho$ is the Hilbert-space pullback of the standard inner product along ρ , while $M_{\text{obs}} = S_+^\dagger S_+$ is the finite-dimensional pullback along S_+ . The quantity that passes between the two levels is the rapidity r .

6.2 The equal-frequency limit

Let $\omega_1 = \omega + \epsilon$, $\omega_2 = \omega - \epsilon$ with $0 < \epsilon < \omega$. Then (10) gives, *exactly*,

$$r = \log \frac{2\omega}{2\epsilon} = \log \frac{\omega}{\epsilon}, \quad d(I, M_{\text{obs}}) = 2 \log \frac{\omega}{\epsilon} \xrightarrow{\epsilon \rightarrow 0^+} \infty. \quad (34)$$

Moreover $\sigma^2 = 4\omega\epsilon$ and (11) give the exact spectral form

$$S_+ = \sqrt{\frac{\omega}{\epsilon}} \Pi_+ + \sqrt{\frac{\epsilon}{\omega}} \Pi_-, \quad (35)$$

so the diagonalizer diverges like $\epsilon^{-1/2}$ while collapsing onto the rank-two spectral projector Π_+ ; the similarity transformation degenerates precisely at the exceptional point $\omega_1 = \omega_2$, where Proposition 2.1 shows the flow eigenvalues merge and A acquires Jordan blocks.

Proposition 6.5 (No scalar renormalization). *For every function $c(\epsilon) > 0$, the spectrum of cM_{obs} is $\{c\omega/\epsilon (\times 2), c\epsilon/\omega (\times 2)\}$, with condition number $(\omega/\epsilon)^2 \rightarrow \infty$. Hence cM_{obs} has no finite, positive-definite, nondegenerate limit as $\epsilon \rightarrow 0^+$: keeping the top eigenvalue finite forces the bottom one to zero.*

The exceptional point of the PU oscillator therefore lies at infinite affine-invariant distance from the identity in the metric cone, and the divergence rate is universal: $d = 2 \log(\omega/\epsilon)$ with no model-dependent prefactor. This is the geometric expression of the known breakdown of the Bender–Mannheim construction at equal frequencies, where $\alpha, \beta \rightarrow \infty$ [4].

7 The Jordan obstruction

At $\omega_1 = \omega_2$ the flow is no longer diagonalizable, and the question becomes whether *any* positive metric can exist for the limiting dynamics. The following finite-dimensional theorem, applied to each Jordan chain, answers it negatively. Parts (1)–(3) for $n = 2$ are fully machine-checked in Lean.

Theorem 7.1 (Jordan no-go). *Let $\omega \in \mathbb{R}$ and $H_J = \begin{pmatrix} \omega & 1 \\ 0 & \omega \end{pmatrix}$, and let $\eta = \eta^\dagger \in M_2(\mathbb{C})$ satisfy the invariance relation $H_J^\dagger \eta = \eta H_J$. Then*

$$\eta = \begin{pmatrix} 0 & b \\ b & d \end{pmatrix}, \quad b, d \in \mathbb{R}, \quad (36)$$

and consequently:

1. no positive-definite invariant Hermitian form exists ($\eta_{11} = \langle e_1, \eta e_1 \rangle = 0$);
2. every positive-semidefinite invariant form is degenerate ($\det \eta = -b^2 \geq 0$ forces $b = 0$);
3. every nondegenerate invariant form is indefinite: for $b \neq 0$ and $\varepsilon = (1 + |d|)^{-1}$ the vectors $x_\pm = (1, \pm \varepsilon b)^\top$ satisfy $\langle x_+, \eta x_+ \rangle > 0 > \langle x_-, \eta x_- \rangle$.

For the $n \times n$ Jordan block, every invariant Hermitian form is anti-triangular Hankel—constant on anti-diagonals and vanishing above the main anti-diagonal—so $\eta_{11} = 0$ and the same trichotomy holds for all $n \geq 2$.

Proof. Writing $H_J = \omega I + N$ with N the nilpotent shift and ω real, invariance reduces to $N^\top \eta = \eta N$, i.e. $\eta_{j-1,k} = \eta_{j,k-1}$ with the boundary convention $\eta_{0,k} = 0$: η is Hankel with $\eta_{1,k} = 0$ for $k \leq n - 1$. For $n = 2$ this gives $\eta_{11} = 0$ and $\eta_{12} = \eta_{21}$, which with Hermiticity makes $b = \eta_{12}$ real. (1) follows from $\eta_{11} = 0$; (2) from $\det \eta = -b^2$ together with positive semidefiniteness of the principal minors; (3) from $\langle x_\pm, \eta x_\pm \rangle = \varepsilon b^2(\pm 2 + \varepsilon d)$ and $\varepsilon|d| < 1$. The general- n statement follows from the Hankel structure and $\langle e_1, \eta e_1 \rangle = \eta_{11} = 0$. $\square$

This is consistent with the general theory of invariant Hermitian forms for matrices with Jordan structure [18] and with the physics of exceptional points [19, 20]: at $\omega_1 = \omega_2$ the PU oscillator admits no positive nondegenerate invariant metric, degenerate or indefinite forms being the only invariant structures. Together with Proposition 6.5, the equal-frequency limit is seen from inside the metric cone as a boundary point at infinite distance, and from the limiting dynamics as an empty cone.

8 Discussion

Relation to Bender–Mannheim. The operator Q , its coefficients α, β , and the positive Hamiltonian $\tilde{H}$ agree exactly with [3, 4]. What is added here is, first, the logical inversion: Q is not an ansatz whose coefficients are fixed by demanding Hermiticity of the transformed Hamiltonian, but the metaplectic logarithm of a diagonalizer that is unique once positivity and the symplectic structure are imposed (Theorems 3.5 and 4.1). Second, the precise positivity statement: the diagonalizer is Hermitian positive only after the canonical normalization $d_x d_y = \gamma \omega_1 \omega_2$ (Proposition 3.3), a point that is easy to miss when working with the operator form alone. Third, the solution family, its stabilizer $SO(2, \mathbb{C})^2$, and the non-invariance of the polar factor (Proposition 3.6) delimit exactly how much of the construction is canonical.

Relation to complex canonical transformations and metric non-uniqueness. Décor, Morales-Técotl, Urrutia and Vergara [10] remove the PU ghost by a complex canonical transformation to two ordinary oscillators, and Mostafazadeh [9] by an alternative Hamiltonian formulation; the transformation S_+ belongs to the same circle of ideas at the level of existence. The classification of Theorem 5.2 is the PU instance of the general non-uniqueness of metric operators in pseudo-Hermitian quantum mechanics [6, 8]; see [15] for a recent discussion of the physical consequences of metric freedom, and [13, 14] for the time-dependent Dyson-map [12] setting in which such maps are studied dynamically. The contribution of Sections 3–5 is to make both the freedom (the stabilizer, the W -family) and the canonical choice (unique positive symplectic diagonalizer) exact and finite-dimensional for this model.

Geometric interpretation. The affine-invariant distance $2r = 2 \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}$ is an intrinsic measure of how non-Hermitian the PU oscillator is: it is the geodesic length, in the natural geometry of the positive cone [17], separating the canonical observable-space metric from the identity, and it depends only on the frequency ratio. Its exact logarithmic divergence, the $\epsilon^{-1/2}$ collapse of S_+ onto a rank-two projector, and the emptiness of the invariant positive cone at the Jordan point (Theorem 7.1) give the equal-frequency exceptional point a complete geometric characterization at the quadratic level.

Possible extensions. The construction used only the data (G, J, G_0) and positivity, and every step—normal form, stabilizer, logarithm, metric cone—is available for any finite set of PT-paired modes. The mode structure of fourth-order field theories, where each spatial momentum contributes a PU pair with $\epsilon(k)$ playing the role of the frequency splitting, suggests a possible extension; we do not pursue it here.

Verification companion. All finite-dimensional algebraic identities appearing in this paper have been independently verified symbolically, numerically, and by machine-checked Lean proofs. The accompanying repository contains the complete verification pipeline: a symbolic audit of every identity (including the derivations of (21) and (30)), a high-precision numerical regression suite covering generic and near-degenerate frequencies, and a Lean 4 development (Mathlib) in which $K^2 = -r^2 I$, $S^\top JS = J$, $\det S = 1$, the congruence $S^\top GS = G_0$, and Theorem 7.1(1)–(3) are formalized without unproved placeholders. We do not claim formal verification of the operator-theoretic statements of Section 5, which remain at the standard formal-algebraic level.

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